

1 **Treatment Limitation Documentation and Risk**
2 **Adjusted Mortality Benchmarking in 190 US**
3 **Intensive Care Units**

4 Author One¹ and Author Two¹

5 ¹Affiliation to be completed

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Abstract

Rationale. Standardized mortality ratios derived from severity of illness scores are used to compare intensive care unit performance. These scores estimate the probability of death under full treatment and contain no term for treatment limitation. Units differ in how often such limitations are documented.

Objective. To determine whether between unit variation in treatment limitation documentation reflects patient case mix or local practice, and to quantify its consequences for risk adjusted mortality benchmarking.

Methods. Retrospective analysis of 136,236 adult admissions to 190 US intensive care units in the eICU Collaborative Research Database during 2014 and 2015. Treatment limitation documented within 24 hours of admission was classified on an ordered scale from full therapy to comfort measures only. We assessed APACHE IVa calibration within limitation strata. We compared patient characteristics between units in the highest and lowest quintile of limitation rate before and after inverse probability weighting. We then ranked units by standardized mortality ratio with and without limitation status in the risk model.

Measurements and Main Results. Code status was documented in 96.1% of admissions. A limitation beyond full therapy was present in 9.9%. The APACHE IVa calibration slope was 1.005 with an intercept of -0.425 . The standardized mortality ratio rose from 0.62 under full therapy to 1.87 under comfort measures only. The integrated calibration index was 0.342 in the comfort measures stratum and 0.037 to 0.057 elsewhere. After weighting until all 13 measured covariates balanced, units in the highest limitation quintile still documented limitation 2.89 times as often as units in the lowest quintile, compared with 3.58 times before weighting. Adding limitation status to the risk model changed quartile assignment for 23.5% of units and reversed mortality outlier classification for 8.0%. The direction of movement was predicted by a unit's limitation rate with a Spearman correlation of -0.689 . Excluding the 878 admissions under comfort measures orders, six in every thousand, reclassified 11.1% of units.

Conclusions. Units differ severalfold in how often treatment limitation is documented and patient case mix does not account for it. Risk adjusted mortality benchmarking that ignores these decisions systematically disadvantages units that document limitations more often.

Introduction

End of life practice varies widely between intensive care units. Withdrawal of life sustaining treatment among patients who die ranges from 0 to 84% across published cohorts [1]. The Ethicus-2 investigators found comparable variation across 199 units in 36 countries [2]. Variation has been documented between countries, between regions and between units within a single country [1, 3, 4]. Two academic centers at opposite extremes of treatment intensity differed in decision making norms rather than in patients [5]. Hospital level measures of end of life treatment intensity have been developed and validated for precisely this reason [6]. Hart and colleagues studied 277,693 admissions to 141 US units and found the proportion admitted with existing treatment limitations ranged from under 1% to 20.9% after risk adjustment [7]. Measurable center characteristics did not explain that variation [7].

Treatment limitation is common in critical illness. Do not resuscitate orders were recorded in 13% of 24,790 admissions to one tertiary unit over 19 years [8], and in 36.6% of a medical intensive care cohort when orders for life sustaining treatment were included [9]. Timing varies, with roughly a quarter of orders written within 48 hours of admission and the remainder later [10]. These orders carry substantial prognostic weight. Mortality among patients with such orders is several times that of patients without them [9, 11], and the association persists after propensity matching [12]. Placing an order also changes subsequent care, including the intensity of interventions delivered [13].

Severity of illness scores estimate the probability of death under full treatment. The APACHE method was introduced in 1985 and validated on 5815 admissions from 13 hospitals [14]. The APACHE IVa model was derived from 110,558 admissions to 104 units during 2002 and 2003 [15]. Its covariates describe physiology, age, comorbidity and admission diagnosis [15]. Comparable models including SAPS 3 share this structure [16, 17]. No covariate in any of them represents a decision to withhold resuscitation. An order not to resuscitate alters the probability of death without altering any variable the model observes.

The performance of these models is known to be unstable. Discrimination and calibration deteriorate as practice changes, which is why successive versions have been required [18, 19]. Performance also differs across ethnic groups within the same health system [20]. Calibration

is generally the first property to fail while discrimination is preserved [18].

Standardized mortality ratios derived from these scores are the accepted currency of intensive care benchmarking [21]. Units are compared and ranked on observed deaths divided by predicted deaths, and in some systems reimbursed on that basis [21]. The ratio is assumed to isolate quality of care once case mix is accounted for [21]. Several lines of evidence suggest the assumption is fragile. Glance and colleagues showed that the choice of scoring system determines which units are labeled outliers [21]. Kramer and colleagues compared APACHE IVa against a National Quality Forum model across 89,353 admissions and found the two agreed on the direction and significance of the standardized mortality ratio in only 45% of units [22]. Admission patterns alone alter mortality based performance measures [23]. Risk adjusted mortality carries conceptual limits that constrain its use as a quality signal, since a statistically unexpected death is not the same as a preventable one [24]. Model misspecification introduces bias into the resulting quality measures [25], and alternative modeling strategies produce materially different unit level estimates [26, 27].

Whether treatment limitation practice distorts these comparisons has not been tested. Hart and colleagues proposed that advance care planning norms contribute to variation in intensive care use and demonstrated the mechanism by simulation [28]. Feng and Dubin identified variables associated with incorrect APACHE IVa predictions in a multicenter database and included the frequency of care limitation change among them, but examined classification error rather than calibration [29]. No study has measured the consequence for risk adjusted mortality benchmarking in observed data.

We examined 136,236 admissions to 190 US intensive care units. We asked whether APACHE IVa calibration depends on documented treatment limitation, whether between unit variation in limitation survives balancing of patient case mix, and whether accounting for limitation changes how units are ranked.

Methods

Data source

We analyzed the eICU Collaborative Research Database version 2.0, which contains 200,859 unit stays at 208 US hospitals participating in a tele critical care program during 2014 and 2015 [30]. The database carries vital signs, laboratory values, care plan documentation, severity scores and outcomes [30], and has been characterized in detail elsewhere [31]. It is deidentified and was accessed under a data use agreement following completion of human subjects research training [30]. Institutional review board approval was waived for analysis of this deidentified resource. Reporting follows the TRIPOD recommendations for studies evaluating prediction models [32, 33].

Cohort

We included adult unit stays with a valid APACHE IVa predicted hospital mortality linked to the APACHE predictor table, yielding 136,236 stays at 190 units. Unit level analyses were restricted to the 162 units contributing 100 or more stays, comprising 135,325 admissions. Figure 1 shows the derivation.

Exposure

Treatment limitation was taken from structured care plan documentation. For each stay we recorded the most restrictive value documented within 24 hours of unit admission on an ordered scale. Full therapy formed the reference. Limitation of resuscitation type comprised do not resuscitate, no cardiopulmonary resuscitation, no intubation and no cardioversion. Partial withdrawal comprised no vasopressors or inotropes, no augmentation of care, no blood products and no blood draws. Comfort measures only formed the most restrictive category. Stays with no entry were retained as an undocumented category and reported but not interpreted. The 24 hour window follows prior work distinguishing early from late limitation orders [10] and limits the degree to which the exposure reflects deterioration occurring later in the stay.

Outcome

In hospital mortality was compared against APACHE IVa predicted hospital mortality.

Statistical analysis

Calibration was assessed by logistic recalibration, regressing observed mortality on the logit of predicted mortality. A slope of 1 and an intercept of 0 indicate perfect calibration. Within strata we report the standardized mortality ratio and the integrated calibration index, the mean absolute difference between a smoothed observed risk and the predicted risk [34, 35]. Standard errors were clustered on unit throughout.

The independent contribution of limitation status was estimated by adding it to a recalibration model containing the APACHE linear predictor. Malignancy was included to test whether an established source of miscalibration behaves similarly [36]. Discrimination was assessed on a random 30% holdout of 40,871 admissions by area under the receiver operating characteristic curve with DeLong's test and by Brier score.

To distinguish practice from case mix we compared units in the highest and lowest quintile of limitation rate. Patient characteristics were balanced by inverse probability of treatment weighting estimated by logistic regression, following recommended practice [37]. Balance was assessed by standardized mean differences against a threshold of 0.1 [37]. We then compared limitation rates between quintiles before and after weighting. If case mix accounted for the difference, balancing case mix should remove it. Between unit variation was additionally summarized by the intraclass correlation coefficient and the median odds ratio from mixed effects models with a unit random intercept [38, 39].

To quantify the benchmarking consequence, expected deaths per unit were computed twice. The first used a recalibrated APACHE IVa model representing current practice. The second added limitation status. Units were ranked by standardized mortality ratio under each and compared on rank shift, quartile change and outlier status using exact Poisson 95% limits following Spiegelhalter [40]. We additionally evaluated two simpler strategies, excluding admissions under comfort measures orders and excluding all admissions with any limitation. Analyses used R version 4.6.0.

Results

Cohort

Among 136,236 admissions, mean APACHE IVa predicted mortality was 11.8% against observed mortality of 8.85%, giving an overall standardized mortality ratio of 0.75. Code status was documented within 24 hours in 96.1% of admissions. A limitation beyond full therapy was present in 13,496 admissions, or 9.9%. Table 1 shows cohort characteristics by limitation status. Patients with any limitation were older, with a median age of 77 years in the resuscitation limitation group against 63 years under full therapy. They were more severely ill, with a median APACHE IVa score of 66 against 49. Patients under comfort measures orders had a median APACHE IVa score of 85 and a median unit stay of 18.2 hours.

APACHE IVa calibration depends on treatment limitation

The overall calibration slope was 1.005 with an intercept of -0.425 . The model ranks patients almost perfectly and is uniformly displaced, over predicting death across the entire risk range. This pattern of preserved discrimination with failed calibration matches that described for severity scores in contemporary cohorts [18].

Calibration differed sharply by limitation status (Figure 2A, Table 2). The standardized mortality ratio was 0.62 under full therapy, 1.17 with resuscitation limitation, 1.22 with partial withdrawal and 1.87 under comfort measures only. The integrated calibration index was 0.039, 0.037 and 0.057 in the first three strata and 0.342 under comfort measures. In that stratum the model is not miscalibrated but uninformative.

In a recalibration model with full therapy as reference, the adjusted odds of death were 2.63 with resuscitation limitation, 2.97 with partial withdrawal and 15.30 under comfort measures only (Figure 2B). Metastatic or hematologic malignancy carried an adjusted odds ratio of 1.15. Adding limitation status reduced model deviance by 3.6% against 0.0% for malignancy, and the likelihood ratio statistic was 2136 against 17.

On held out data, adding limitation status improved the area under the curve from 0.8657 to 0.8760, significant by DeLong's test at $p < 0.001$, and reduced the Brier score from 0.0621 to

0.0579 (Table 3). Adding malignancy changed the area under the curve by 0.0001, which was not significant.

Variation between units is not explained by case mix

Across 162 units, limitation rates ranged from 0 to 27.5% with a median of 10.5% (Figure 3A). One unit documented no limitation across more than 100 admissions.

Units in the highest limitation quintile differed from those in the lowest on two of 13 measured characteristics before weighting. Patients were older, with a standardized mean difference of 0.197, and more often admitted from the emergency department, at 0.149. Severity was near identical, at -0.006 for the APACHE IVa score and 0.008 for predicted mortality.

After inverse probability weighting all 13 covariates balanced within the 0.1 threshold (Figure 3B). Age fell to -0.004 and admission from the emergency department to 0.004. Despite complete balance on measured characteristics, units in the highest quintile documented limitation in 16.5% of admissions against 5.7% in the lowest, a ratio of 2.89. The unweighted ratio was 3.58. Approximately four fifths of the difference in practice survived balancing of patient case mix.

Mixed effects models gave the same answer. The intraclass correlation was 0.084 before adjustment and 0.104 after adjustment for case mix. The median odds ratio rose from 1.69 to 1.80, indicating that moving a patient between randomly chosen units changes the odds of documented limitation by close to twofold [39].

Consequences for benchmarking

Adding limitation status to the risk model preserved the overall ordering of units, with a Spearman correlation of 0.949, while moving individual units substantially (Table 4). The median absolute rank shift was 7.5 positions of 162. Sixty two units moved more than 10 positions and 24 moved more than 20.

Thirty eight units, or 23.5%, changed quartile. Thirteen units, or 8.0%, changed outlier classification against exact Poisson limits [40]. Four units previously flagged as high mortality became unremarkable and three previously unremarkable units became outliers (Figure 4B).

Movement was systematic. The signed rank shift correlated with a unit's limitation rate at a Spearman coefficient of -0.689 (Figure 4A). Units in the lowest limitation quintile moved from a standardized mortality ratio of 0.984 to 1.027. Units in the highest quintile moved from 1.013 to 0.876. Under current benchmarking these quintiles appear equivalent. After accounting for limitation they do not.

A simpler remedy

Excluding the 878 admissions under comfort measures orders, 0.6% of the cohort, reclassified 18 units across quartiles, or 11.1% (Table 5). Full adjustment for the ordered limitation scale reclassified 38 units, or 23.5%. Excluding all 13,496 admissions with any limitation reclassified 46 units, or 28.4%. Excluding the smallest group recovered roughly half the effect of full adjustment.

Discussion

APACHE IVa calibration depends on a decision the model does not observe. The standardized mortality ratio was 0.62 among patients receiving full therapy and 1.87 among patients under comfort measures. The integrated calibration index in that stratum was nine times larger than elsewhere [34]. The model is not merely displaced in these patients. It is uninformative.

Documentation of limitation varied severalfold between units and patient case mix did not account for it. This extends Hart and colleagues, who reported comparable variation after regression adjustment in a different database a decade earlier [7]. Our design equalized the covariate distributions rather than adjusting for them [37], and roughly four fifths of the practice difference persisted. Limitation rates were higher in our cohort than in theirs, 10.5% against 4.0%, consistent with the expansion of palliative care services over the intervening period. Palliative care consultation more than doubles transition to limitation orders in randomized comparison, from 23.4% to 50.5% [41], and integration of palliative care into critical care alters both process and outcome measures [42, 43]. Units therefore differ in limitation rate partly because they differ in service configuration, which is a property of the unit rather than of its

patients [5].

The consequence for benchmarking is substantial. Accounting for limitation moved 23.5% of units across quartiles and reversed outlier classification for 8.0%. Kramer and colleagues found that substituting an entirely different risk model changed the direction and significance of the standardized mortality ratio in 55% of units [22]. Glance and colleagues reached a similar conclusion comparing scoring systems [21], and admission patterns alone have been shown to shift these measures [23]. We find that a single variable already recorded in 96% of charts moves a quarter of quartile assignments. The intervention is far smaller and the effect is of comparable order.

Whether to adjust for limitation status is a genuine question and we do not resolve it. Adjustment treats limitation as patient preference and therefore as case mix. Withholding adjustment treats it as modifiable unit behavior. Both positions are defensible. Nerenz and colleagues framed the identical dilemma for social risk factors, where adjustment may conceal real differences in care while omission penalizes institutions serving different populations [44]. The parallel is exact. Misspecification of the adjustment model is itself a recognized source of bias in quality measurement [25]. Our contribution is not to settle the question but to show that the answer changes the rank of one unit in four, and that the choice is currently made by default.

A simpler remedy exists. Excluding admissions under comfort measures orders affects six patients in every thousand and recovers half the effect. Benchmarking programs already exclude categories of patient from severity scoring, and the National Quality Forum model excludes 27.9% of admissions against 10.6% for APACHE IVa [22]. Adding one narrow exclusion is feasible within existing practice.

This study has limitations. We observed documentation and not practice. Code status documentation in electronic records is frequently incomplete, difficult to locate or internally contradictory [45]. Units may limit treatment without recording it, which would understate true variation. The data describe 2014 and 2015 and derive from units participating in a single tele critical care program [30]. Limitation status is partly a consequence of prognosis. Clinicians write these orders because they expect death, so limitation encodes information the score lacks and no causal interpretation is available. This same dependency has been shown to bias

mortality based effect estimates in trauma [46]. Adjusting for a variable partly under unit control carries the hazards of mediator adjustment. Weighting balanced measured characteristics only and unmeasured differences between units may remain [37]. Cancer ascertainment relied on APACHE comorbidity flags and captured metastatic solid tumor and hematologic malignancy only [36].

Intensive care units are ranked on mortality. Mortality depends on whether treatment is given. Whether treatment is given depends on a decision that varies severalfold between units for reasons unrelated to patients. Benchmarking that ignores this decision measures end of life practice alongside quality of care.

Data availability

The eICU Collaborative Research Database is available to credentialed investigators from PhysioNet. Analysis code is available at [repository URL].

Funding

[To be completed.]

Conflicts of interest

[To be completed.]

Tables

Table 1: Characteristics of 136,236 intensive care admissions by treatment limitation status within 24 hours.

Characteristic	Full therapy	DNR type	Partial withdrawal	Comfort measures	Undocumented
Stays, n	117,377	11,929	689	878	5363
Percent of cohort	86.2	8.8	0.5	0.6	3.9
Age, median years	63	77	76	74	64
Age over 89, %	2.2	15.6	19.0	13.8	3.9
Female, %	45.1	54.1	55.9	50.5	45.0
APACHE IVa score, median	49	66	69	85	50
Predicted mortality, %	10.5	22.3	26.2	39.6	11.2
Observed mortality, %	6.5	26.1	31.9	74.1	8.8
Standardized mortality ratio	0.62	1.17	1.22	1.87	0.78
Malignancy, %	2.9	6.1	7.0	6.8	1.9
Immunosuppression, %	2.5	4.3	2.8	6.0	1.6
Ventilated day 1, %	23.6	29.3	32.5	45.8	19.3
ICU stay, median hours	43.2	43.1	36.3	18.2	43.1

APACHE denotes Acute Physiology and Chronic Health Evaluation. Malignancy denotes metastatic solid tumor or hematologic malignancy.

Table 2: APACHE IVa calibration by treatment limitation status.

Code status	n	Predicted, %	Observed, %	SMR	ICI	$E_{\max}$
Full therapy	117,377	10.5	6.5	0.62	0.039	0.241
DNR type	11,929	22.3	26.1	1.17	0.037	0.048
Partial withdrawal	689	26.2	31.9	1.22	0.057	0.078
Comfort measures	878	39.6	74.1	1.87	0.342	0.487
Undocumented	5363	11.2	8.8	0.78	0.022	0.125

SMR denotes standardized mortality ratio. ICI denotes integrated calibration index. $E_{\max}$ denotes the maximum absolute calibration error.

Table 3: Discrimination and accuracy on held out data.

Model	AUC	95% CI	Brier score
APACHE IVa	0.8657	0.8597 to 0.8717	0.0621
APACHE IVa recalibrated	0.8657	0.8597 to 0.8717	0.0605
Plus malignancy	0.8658	0.8599 to 0.8718	0.0605
Plus limitation status	0.8760	0.8701 to 0.8818	0.0579
Plus limitation and malignancy	0.8761	0.8702 to 0.8819	0.0579

Held out sample $n = 40,871$. AUC denotes area under the receiver operating characteristic curve. Adding limitation status improved the AUC ($p < 0.001$, DeLong). Adding malignancy did not ($p = 0.16$).

Table 4: Effect of modeling treatment limitation on unit ranking.

Measure	Value
Spearman correlation of ranks	0.949
Median absolute rank shift	7.5 of 162
Units moving more than 10 positions	62 (38.3%)
Units moving more than 20 positions	24 (14.8%)
Units changing quartile	38 (23.5%)
Units changing outlier classification	13 (8.0%)
High mortality outlier to as expected	4
As expected to high mortality outlier	3

Based on 162 units with 100 or more admissions. Outlier status from exact Poisson 95% limits.

Table 5: Comparison of remediation strategies.

Strategy	Admissions affected	Units changing quartile	Correlation with current
Current practice, APACHE IVa alone	0	reference	1.000
Adjust for limitation status	0	38 (23.5%)	0.949
Exclude comfort measures only	878 (0.6%)	18 (11.1%)	0.986
Exclude any limitation	13,496 (9.9%)	46 (28.4%)	0.878

Correlation is the Spearman rank correlation of standardized mortality ratios against current practice.

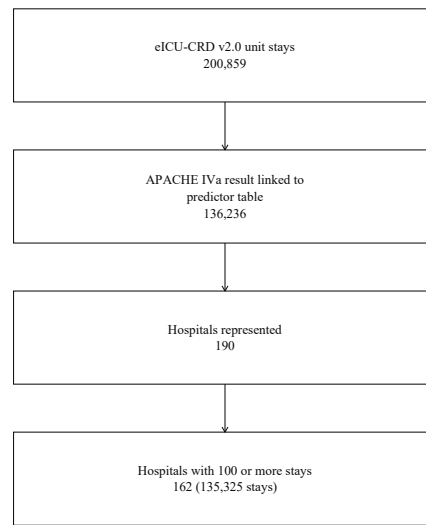


Figure 1: Derivation of the study cohort from the eICU Collaborative Research Database version 2.0.

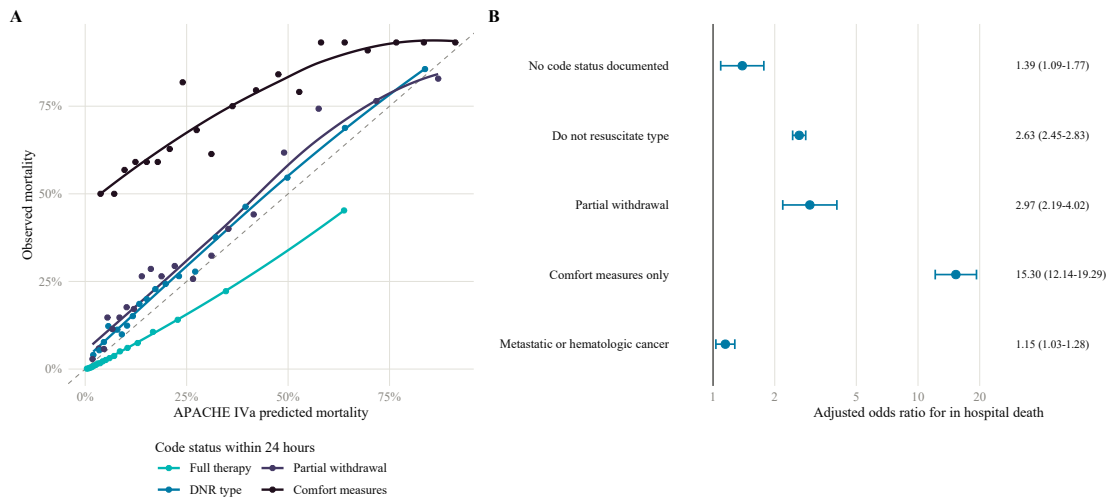


Figure 2: APACHE IVa performance by treatment limitation status. (A) Calibration curves. Points are 5% risk bins and the dashed line indicates perfect calibration. (B) Adjusted odds ratios for in hospital death from a logistic recalibration model, with full therapy as reference and standard errors clustered on unit.

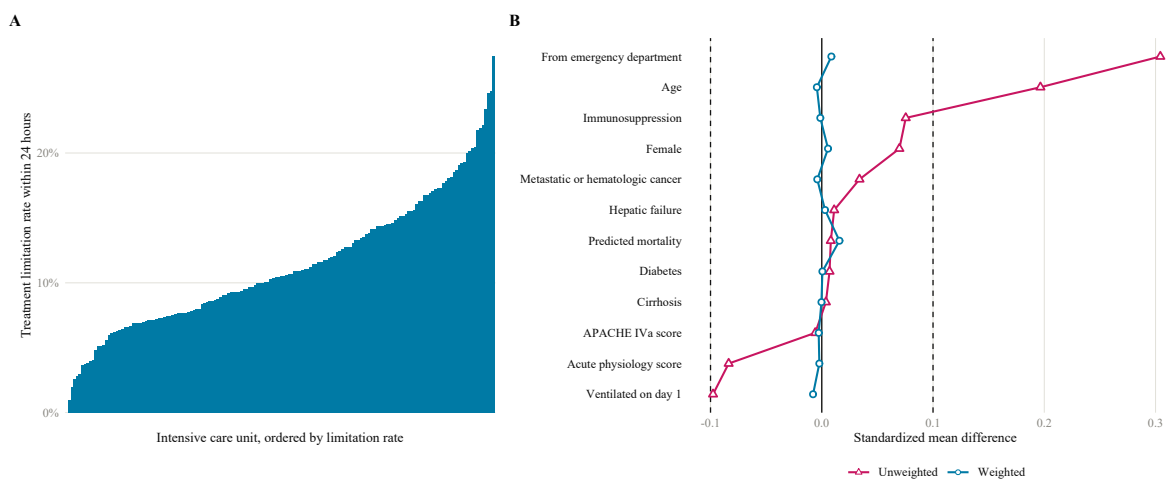


Figure 3: Between unit variation in treatment limitation. (A) Limitation rate within 24 hours across 162 units with 100 or more admissions, ordered. (B) Standardized mean differences in patient characteristics between units in the highest and lowest limitation quintile, before and after inverse probability weighting. The shaded band marks the conventional 0.1 threshold.

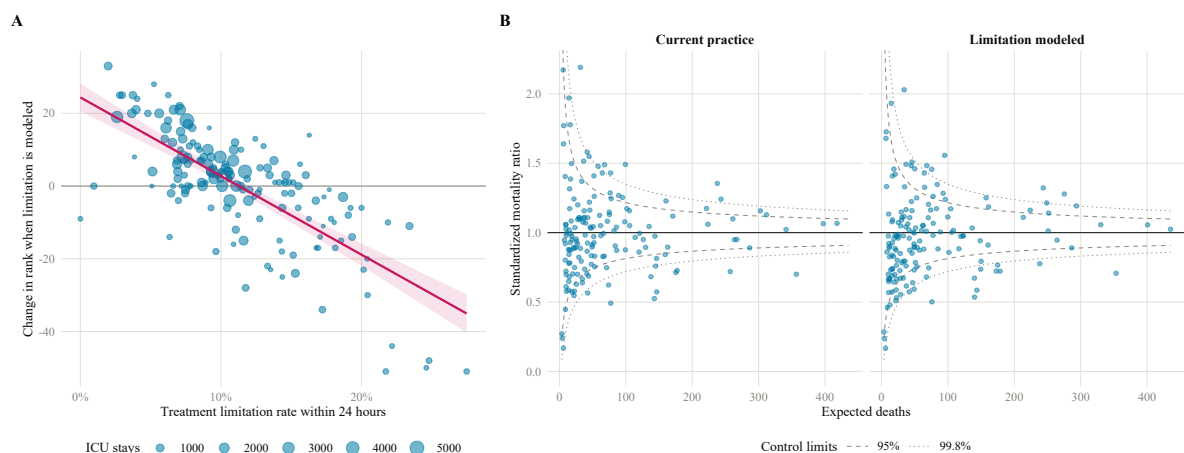


Figure 4: Effect of modeling treatment limitation on unit benchmarking. (A) Change in unit rank against limitation rate. Negative values indicate improvement. (B) Funnel plots of standardized mortality ratio against expected deaths under current practice and with limitation modeled, with 95% and 99.8% exact Poisson control limits.

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